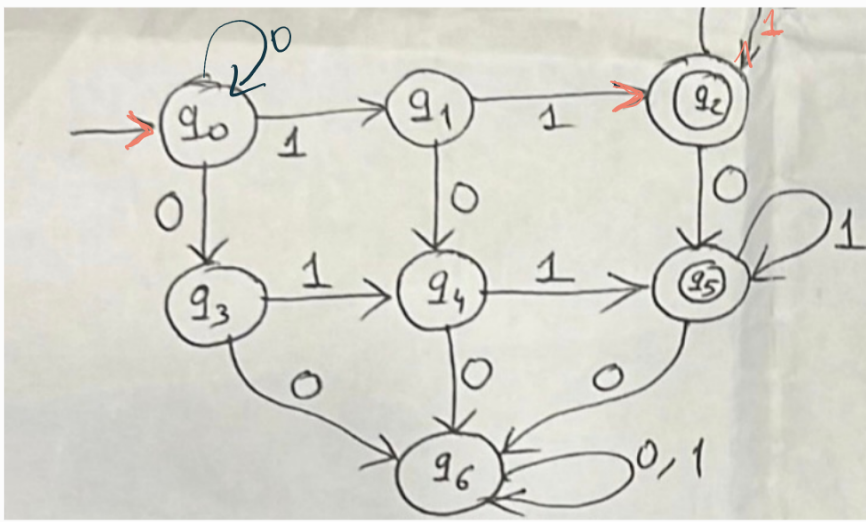




$$\Sigma = \{0, 1\}$$

$$F = \{q_2, q_5\}$$

tanımlı NFA- Λ

Nerden başlıya girdiği yazıldı,

$$q_0 = \Lambda$$

$$q_1 = q_0 1 = 1$$

$$q_2 = q_1 1 + q_2 1 = q_1 1^* = 111^*$$

$$q_3 = q_0 0 = 0$$

$$q_4 = q_1 0 + q_3 1 = 10 + 01$$

$$q_5 = q_4 1 + q_2 0 + q_5 1 = ((10 + 01)1 + 111^* 0)1^*$$

$$q_6 = q_3 0 + q_4 0 + q_5 0 + q_6 1 + q_6 0$$

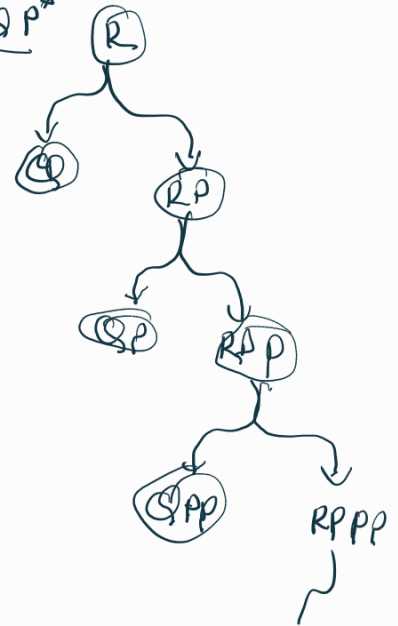
$$F = q_2 + q_5$$

$$= 111^* + ((10 + 01)1 + 111^* 0)1^*$$

Arden teoremi

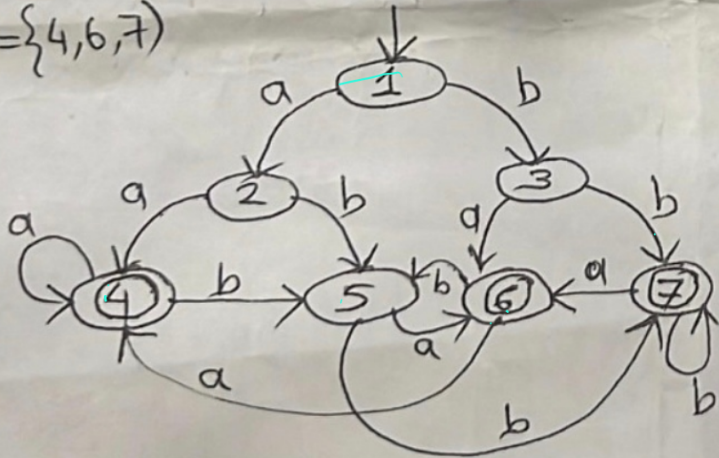
$$R = Q \overset{x}{\oplus} R P^*$$

$$R = Q P^*$$



Soru 5. Aşağıdaki DFA'da durum sayısını minimuma indirgeyiniz.

$F = \{4, 6, 7\}$



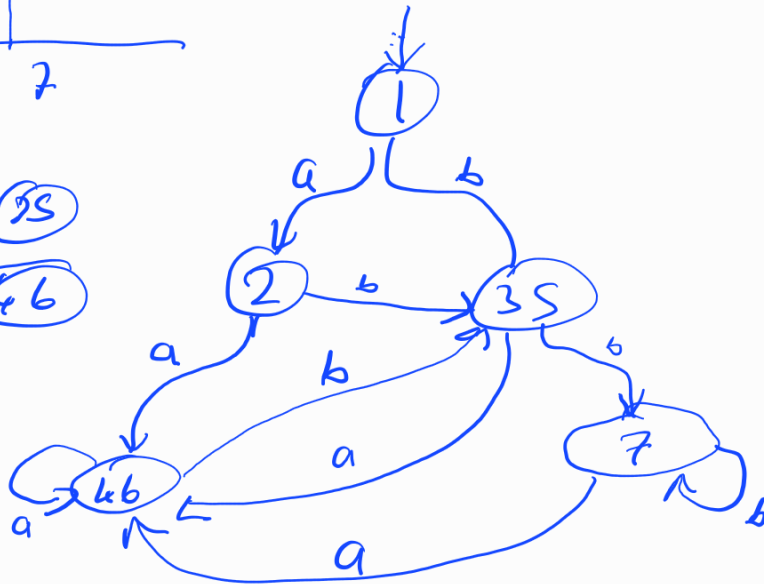
$f =$

1							
2	x						
3	x	x					
4	x	x	x				
5	x	x	v	x			
6	x	x	x	v	x		
7	x	x	x	x	x	x	
	1	2	3	4	5	6	7

λ
a
b
aa
ab
ba
bb

$3 \sim 5$ (35)

$6 \sim 4$ (46)



$A = b$

$B = Aa = a$

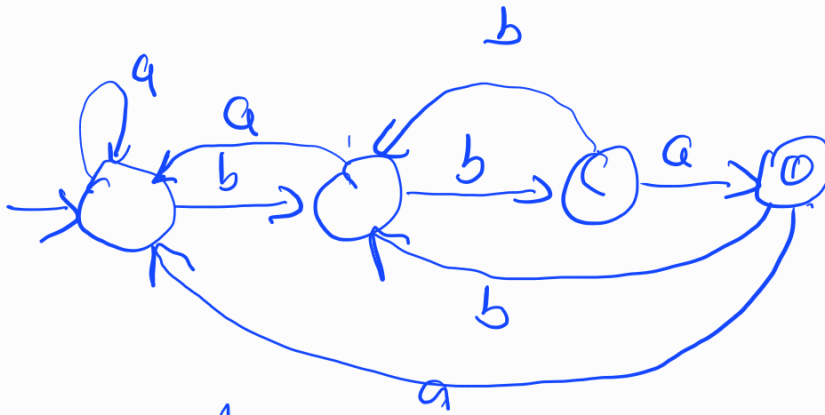
$C = Ab + Bb + Db = b + ab + Db$

$D = Ea + Ca + Ba + Da = (Cbb^*a + Ca + aa)a^*$

$E = Cb + Eb = Cbb^*$

bbq ile biten DFA çiz

1-Soru

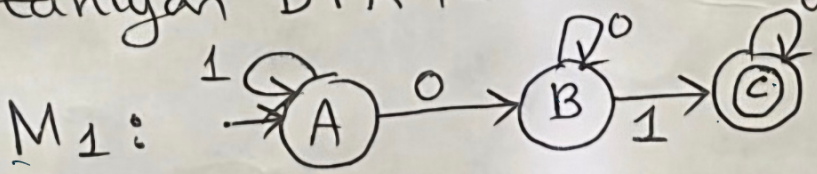


$(a+b)^* bba$

Soru 6. Aşağıdaki makine
hangi dili tanır?

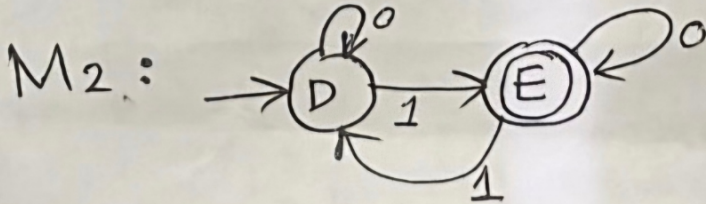
$L(M_1) \cup L(M_2)$, $L(M_1) \cap L(M_2)$

ve $L(M_1) - L(M_2)$ dillerini
tanıyan DFA'ları çiziniz.

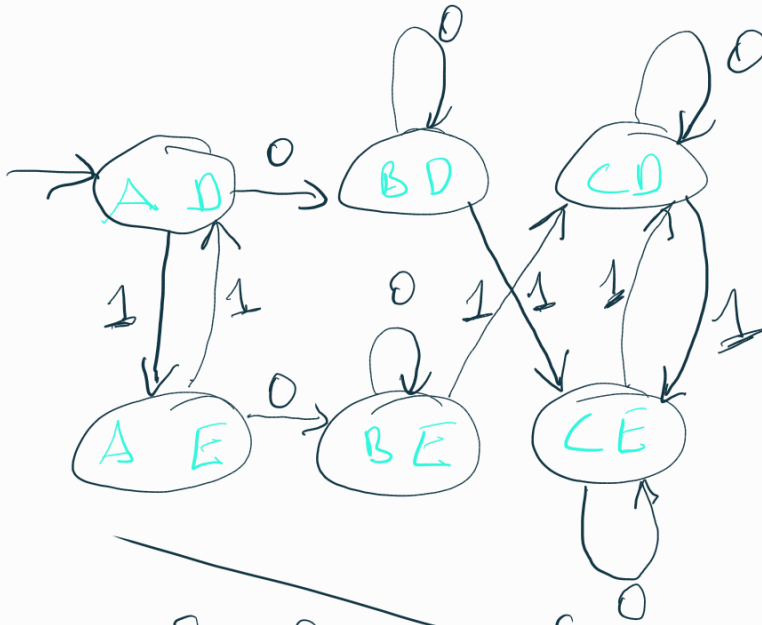


$$f_1 = \{c\}$$

$$f_2 = \{e\}$$



Süre 60 dakika



$$L_1 \cup L_2 = \bigcup_{L} f_2 \times Q_1 \cup Q_2 \times f_1$$

$E \times \{A, B, C\}$

$$L_1 \cap L_2 = f_1 \times f_2$$

$= \underline{CE}$

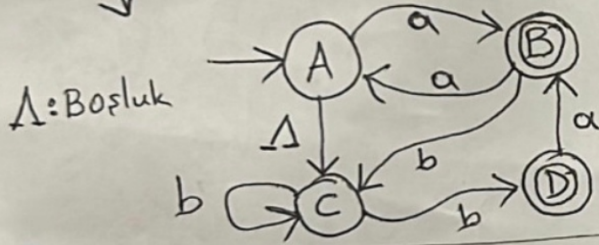
$$(E \times \{A, B, C\}) \cup \{D, E\} \times \{C\}$$

$L_1 - L_2 = \underline{CD}$

(CD, CE)

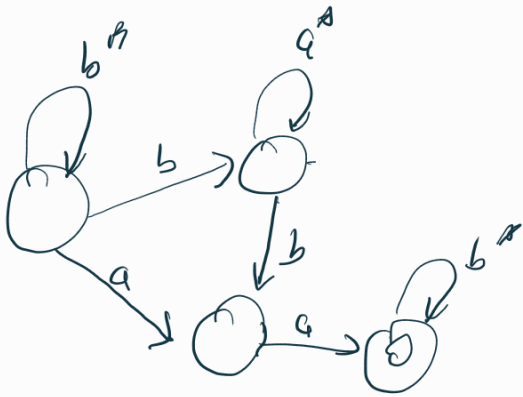
$$L_1 \cup L_2 \rightarrow \boxed{AE, BE, CE, CD}$$

Soru 3. $\Sigma = \{a, b\}$ alfabesinde tanımlı makinesine ilişkin eşdeğer DFA makinesini çiziniz



Soru 4: $L = b^*(ba^*b + a)ab^*$ regüler ifadesi için bir NFA makinesi çiziniz.

4)

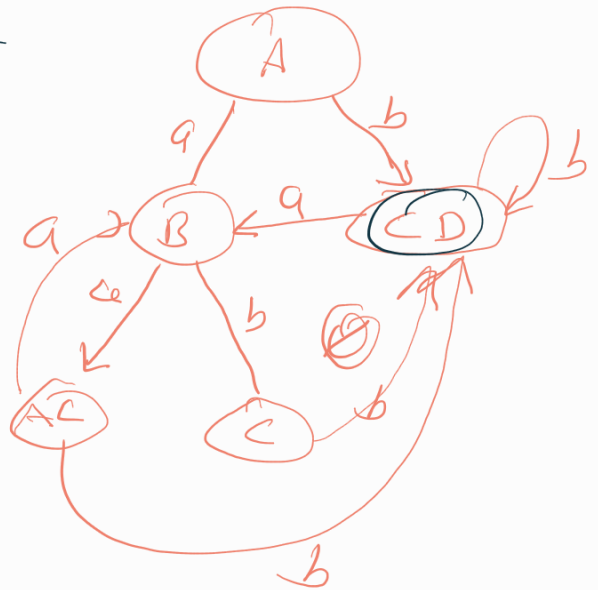


$A \cup C$
 $\underline{B}$
 $\underline{AC}$
 $\underline{BC}$

3)

	$S(q, a)$	$S(q, b)$	$S^*(q, a)$	$S^*(q, b)$
A	C	B	-	-
B	-	A	C	CD
C	-	-	AC	C
D	-	B	-	CD

DFA



NFA:

